

Non-Uniform Linear Arrays for Target Detection and DoA Estimation in Passive Radar STAP

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Abstract— This paper explores the use of Non-Uniform Linear Array (NULA) configurations for target detection and Direction of Arrival (DoA) estimation in mobile passive radar systems. By leveraging on spatial diversity, NULAs are shown to potentially improve the flexibility and performance of a Space-Time Adaptive Processing (STAP) scheme, when few spatial receiving channels are available, as is typically the case of passive radar. The impact of NULAs on clutter suppression performance is analysed, showing that a narrower cancellation notch can be obtained, enhancing the detection of slow-moving targets, while still preventing the comparison of blind velocities. Moreover, resorting to a non-uniform distribution of the antennas, the trade-off between DoA estimation accuracy and unambiguous angular sector can be relaxed. A larger displacement between the phase centres is shown to reduce the DoA estimation error of unambiguous targets. The advantages of the proposed solution are demonstrated against simulated data, addressing the case of a mobile passive radar exploiting DVB-T as a signal of opportunity.

Keywords—Passive radar, STAP, GMTI, NULA, Clutter cancellation, DoA estimation

I. INTRODUCTION

In the last few years, passive radar systems mounted on moving platforms have gained a considerable attention in the scientific community. The growing interest in these systems is posing new challenges in terms of processing techniques, operational strategies, and system design. One of the crucial tasks to be addressed when dealing with mobile passive radars is clutter cancellation. Thanks to the application of space-time processing techniques, clutter suppression can be achieved by imposing a cancellation notch in the angle-Doppler domain, allowing to highly attenuate the returns from the stationary scene. The effectiveness of these techniques for passive radar has been proved against both simulated and experimental data in [1],[2], based on a Displaced Phase Centre Antennas (DPCA) approach, and in [3]-[5], based on a Space-Time Adaptive Processing (STAP) scheme, exploiting FM and DVB-T as illuminators of opportunity and multichannel receivers with Uniform Linear Array (ULA) configurations.

As it is well known, passive radar systems are oriented to the design of low-cost, lightweight, and compact systems. Furthermore, when exploiting signals of opportunity at the VHF/UHF frequencies, the antennas are characterized by a non-negligible size. These requirements lead to a severe limitation in the number of available channels. In addition to that, these systems typically employ low-directive commercial antennas. Therefore, estimating the Direction of Arrival (DoA) of the detected targets represents a difficult task. In [4] and [5], a Maximum Likelihood Estimation (MLE) approach has been adopted to provide simultaneous clutter cancellation, target detection and DoA estimation via an adjacent-bin post-Doppler STAP approach.

In this context, the advantages of employing longer linear arrays in reception is twofold. On the one hand, it could facilitate the detection of very slow-moving target, which would otherwise be attenuated by the cancellation notch. On the other hand, longer arrays could be beneficial to the DoA estimation accuracy, leading to a narrower main beam in the likelihood function of the estimator.

However, if a low, fixed number of receiving elements is available, the only way to increase the total length of the array is by increasing the inter-element spacings. When adopting ULA configurations, this results in a spatial undersampling, which has two main undesirable consequences:

1. a potential folding of the clutter spectrum which may result in the comparison of blind velocities, undermining the detection of relatively faster targets;
2. a decrease in the unambiguous angular sector for target DoA estimation, due to the appearance of grating lobes in the likelihood function.

To address the above issues, this paper suggests the use of Non-Uniform Linear Array (NULA) configurations, as a way to strategically position a limited number of antenna elements. The impact of NULA configurations on target detection and DoA estimation performance of mobile passive radar systems represents an interesting and unexplored perspective. The issues deriving from spatial undersampling could be overcome by capitalizing on the spatial diversity provided by NULAs.

A first study characterizing the rank of the clutter covariance matrix for radar systems based on non-uniform array can be found in [6]. The advantages of NULA configurations in DoA estimation have been studied in [7], for a clutter-free scenario.

In this work, we provide a preliminary assessment of the advantages of a non-uniform distribution of antennas in passive radar GMTI applications, focusing on both target detection and DoA estimation capabilities, and making use of a STAP technique.

The paper is organized as follows. In Section II, we introduce the system geometry, the signal model, and the adopted processing scheme. In Section III, the advantages, and disadvantages of NULA in terms of target detection and DoA estimation are investigated, making use of a simulated data set. Finally, in Section IV, we draw our conclusions.

II. SYSTEM GEOMETRY AND PROCESSING SCHEME

Let us consider a passive radar system mounted on a platform moving at a constant velocity v_p along the direction of the y axis. We assume that the radar receiver is equipped with a linear array of N antennas, aligned in the along-track direction (side-looking geometry). The n -th element position

at the starting time of the acquisition is denoted as d_n . Without loss of generality, we assume that the first element of the array is located at $y = 0$, so that the array configuration is described by:

$$\mathbf{z} = [0, d_1, \dots, d_{N-1}] \quad (1)$$

Figure 1 illustrates the considered system bistatic geometry. We assume that the exploited transmitter of opportunity is ground-based and located behind the receiver. We denote with θ_0 the angle between the array end-fire and the receiver to scatterer line of sight.

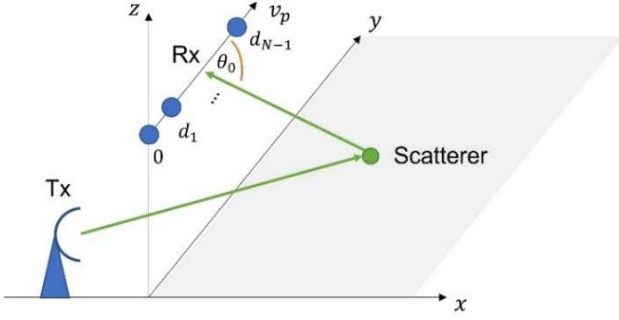


Figure 1– System Geometry.

The observed scene is characterized by the presence of stationary scatterers, which contribute to clutter returns and may hide the echoes from potential moving targets. Due to the relative motion between the receiver and the observed scene, the clutter signal is not concentrated at zero Doppler frequency. Therefore, space-time processing approaches such as DPCA and STAP are needed to achieve clutter cancellation and discriminate the presence of moving targets.

In this work, a STAP approach is considered to achieve clutter cancellation and moving target detection. Specifically, the adopted processing scheme, which was firstly presented in [4], is sketched in Figure 2.

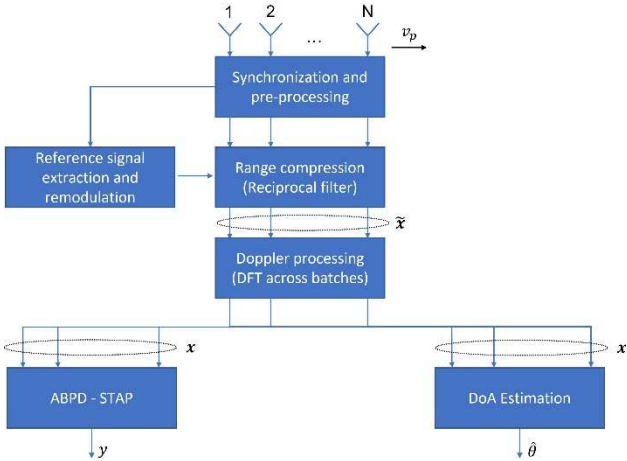


Figure 2 – Sketch of the adopted processing scheme.

A replica of the transmitted signal is assumed to be available, obtained by means of a decode/recode approach. To evaluate the range-Doppler maps, the continuous wave signal is segmented into batches, which allows to emulate the conventional fast-time/slow-time framework typical of a pulsed radar system, operating at an equivalent PRF given by the inverse of the batch length ($PRF = 1/T$). Let M denote

the number of pulses in the Coherent Processing Interval (CPI). The M batches are separately range compressed, and Doppler processed along the equivalent slow time dimension.

The range compression stage exploits a reciprocal filter. As known, the reciprocal filter allows both to improve the ambiguity function and to remove the variability associated to the information content of the waveform of opportunity. This has a key role in enabling an effective clutter suppression in the subsequent stages, as duly described in ([2]).

After range compression, the complex-valued samples received by the N antennas, relative to the M pulses and associated to the range cell under test are arranged in the $(NM \times 1)$ space-time data vector $\tilde{\mathbf{x}}$. The signal received at the m -th pulse and at the n -th antenna is characterized by the following phase contribution:

$$s_{n,m} = e^{j\frac{2\pi}{\lambda}d_n \cos \theta_0} e^{j2\pi f_d m T} \quad (2)$$

The first phase term is due to the path difference between the n -th antenna and the first one, assumed as a reference. The second phase term is the Doppler frequency shift caused by the relative motion between the receiver and the scatterer. Note that the phase shift within each batch has been neglected. The bistatic Doppler frequency f_d can be decomposed into two contributions:

$$f_d = \frac{v_p \cos \theta_0}{\lambda} - \frac{v_b}{\lambda} \quad (3)$$

the first is due to the radial component of the platform velocity, the second to the intrinsic radial velocity of the observed target.

This allows us to define the spatial steering vector as:

$$\mathbf{s}_s(\theta) = \left[1, e^{-j2\pi \frac{d_1}{\lambda} \cos \theta}, \dots, e^{-j2\pi \frac{d_{N-1}}{\lambda} \cos \theta} \right] \quad (4)$$

and the temporal steering vector as:

$$\mathbf{s}_t(f_d) = \left[1, e^{-j2\pi f_d T}, \dots, e^{-j2\pi f_d (M-1)T} \right] \quad (5)$$

so that the space-time steering vector is given by:

$$\mathbf{s}(f_d, \theta) = \mathbf{s}_t(f_d) \otimes \mathbf{s}_s(\theta) \quad (6)$$

As known, STAP techniques adaptively combine spatial and temporal samples of the signal to suppress clutter in the angle-Doppler domain and maximize the probability of detecting moving targets. This is achieved through the estimation and inversion of the disturbance covariance matrix $\mathbf{Q}$, which typically makes STAP techniques computationally intensive. To tackle this issue, several reduced-order STAP techniques were developed, [8]-[9]. In this paper, we resort to the Adjacent-Bin Post-Doppler (ABPD) approach adopted in [4], where adaptive filter is applied on the N spatial samples from a subset of L adjacent Doppler bins around the cell under test.

Let $\mathbf{x}$ ($NL \times 1$) denote the data vector obtained after Doppler processing and adjacent bin selection. This data vector is the input of the ABPD-STAP block in Figure 2, which applies the following space-time GLRT (Generalized Likelihood Ratio Test) detector, derived from [10]:

$$\frac{|s^H \hat{\mathbf{Q}}^{-1} \mathbf{x}|^2}{s^H \hat{\mathbf{Q}}^{-1} s (1 + \mathbf{x}^H \hat{\mathbf{Q}}^{-1} \mathbf{x})} \leq \eta \quad (7)$$

where the disturbance matrix $\mathbf{Q}$ has been substituted by its ML estimate $\hat{\mathbf{Q}} = \mathbf{X}_k \mathbf{X}_k^H$, being $\mathbf{X}_k = [\mathbf{x}_1, \dots, \mathbf{x}_{N-1}]$ a set of training data of size $(NL \times K)$ from K adjacent range cells.

As mentioned in the introduction, after the target detection stage, we are interested in estimating the DoA of the detected targets. Since passive radar are typically characterized by low-directive antennas, the DoA estimation performance are extremely important for a proper target localization. In this paper, we resort to the Maximum Likelihood (ML) estimation technique as in [4].

The log-likelihood function to be maximized with respect to the DoA θ is given by:

$$\hat{\theta}_t = \underset{\theta}{\operatorname{argmax}} \left\{ \frac{|s(\theta)^H \hat{\mathbf{Q}}^{-1} \mathbf{x}|^2}{s(\theta)^H \hat{\mathbf{Q}}^{-1} s(\theta)} \right\} \quad (8)$$

where θ_t represents the true target DoA.

In the next section, the described processing scheme is tested in the case of a NULA configuration, analysing its impact on both target detection and DoA estimation performances, and providing a comparison with the ULA case with the same number of channels. The expected advantage of NULA is twofold: on the one hand, it should enable the detection of very slow-moving target, without causing the appearance of blind velocities. On the other hand, it should be beneficial to the DoA estimation stage, allowing to achieve better accuracy without causing the appearance of ambiguous DoA estimates, which in turn would lead to a decrease in the unambiguous angular sector.

III. TARGET DETECTION AND DOA ESTIMATION PERFORMANCE WITH NULAS

In this section, the advantages and disadvantages of a NULA configuration in terms of target detection and DoA estimation are illustrated. To this aim, the following scenario has been simulated. We consider the case of a ground receiver moving at velocity $v_p = 13 \text{ m/s}$. An 8k mode DVB-T signal sequence is assumed as a reference signal. Clutter returns are generated for a scene spanning $N_R = 1000$ range cells. The complex amplitudes of the scatterers are assumed to be independent and identically distributed, following a gaussian

probability density function. Internal clutter motion is assumed absent.

Figure 3 shows the range-Doppler map obtained at a single channel. Specifically, we assumed a CPI of 0.5s. The simulated signal also includes two targets, located at DoA $\theta_t = 90^\circ$, and moving at different bistatic radial velocities, $v_b^{(1)} = 3 \text{ m/s}$ and $v_b^{(2)} = 13 \text{ m/s}$. Both the targets appear severely masked by the Doppler-spread clutter returns. The injection of targets with different velocities allows us to characterize the behaviour of ULA and NULA configurations.

To this aim, we consider three array configurations with $N = 3$ receiving elements, namely the $\lambda/2$ spaced ULA $\mathbf{z}_1 = [0 \ 0.5 \ 1]\lambda$, the λ spaced ULA $\mathbf{z}_2 = [0 \ 1 \ 2]\lambda$ and the NULA $\mathbf{z}_3 = [0 \ 0.5 \ 2]\lambda$. Notice that the NULA configuration has been arbitrarily chosen based on the optimal design strategies proposed in [11].

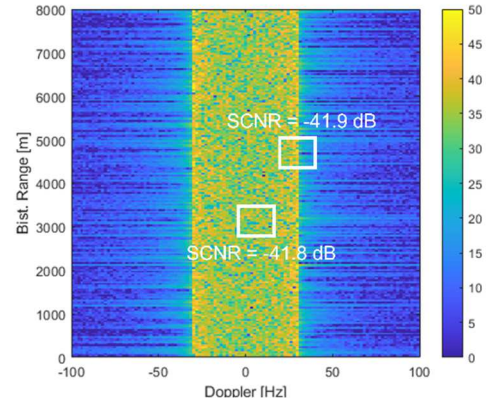


Figure 3 – Single channel range-Doppler map.

A. Target Detection

Figure 4 (a), (b) and (c) show the three range-Doppler maps obtained after the ABPD-STAP block with $L = 3$ Doppler bins and $K = 24$ secondary data, using the three array configurations defined above. These figures provide a qualitative overview of the advantages of NULAs in terms of target detection. In particular, we notice that:

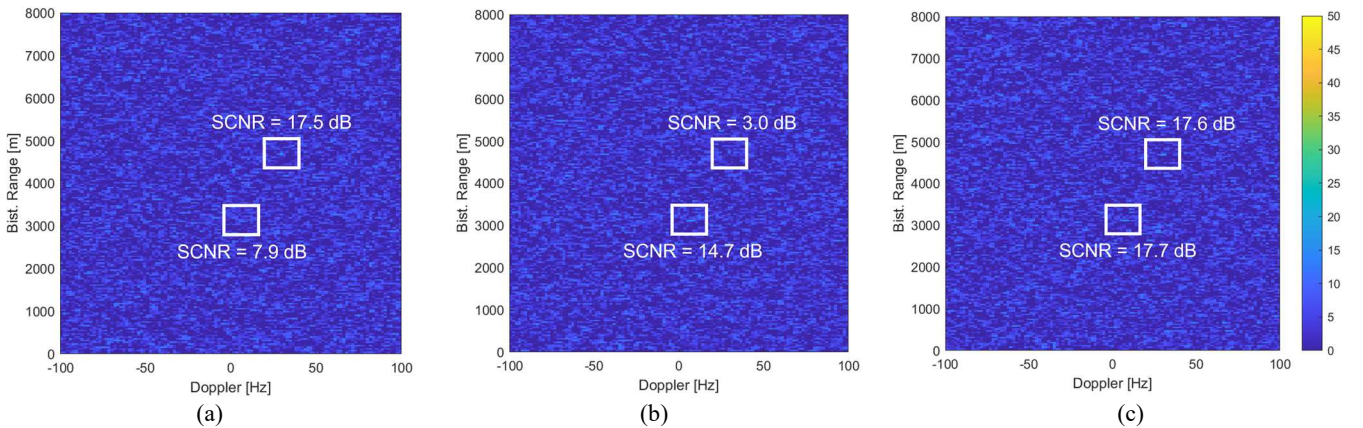


Figure 4 – Range-Doppler maps obtained using three different array configurations. (a): ULA $\mathbf{z}_1 = [0 \ 0.5 \ 1]\lambda$. (b): ULA $\mathbf{z}_2 = [0 \ 1 \ 2]\lambda$. (c): NULA $\mathbf{z}_3 = [0 \ 0.5 \ 2]\lambda$.

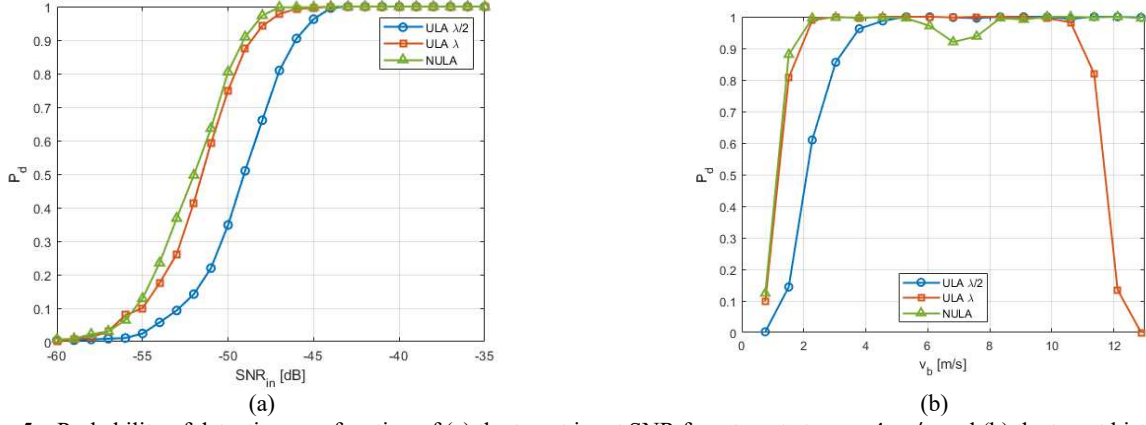


Figure 5 – Probability of detection as a function of (a) the target input SNR for a target at $v_b = 4 \text{ m/s}$ and (b) the target bistatic target velocity for a fixed SNR of -45 dB .

The target at $v_b^{(1)} = 4 \text{ m/s}$ is highly attenuated due to the wider cancellation notch obtained using the $\lambda/2$ -ULA. On the other hand, the target at $v_b^{(1)} = 13 \text{ m/s}$ is almost undetectable when using the λ -ULA, due to the presence of a replica of the cancellation notch, caused by spatial undersampling. Finally, when using the NULA, both the targets are preserved, as the cancellation notch is narrower than the one yielded by the $\lambda/2$ -ULA, while the blind velocities affecting the λ -ULA have been prevented.

To numerically assess the target detection performance, we focused on a slow-moving target with $v_b = 4 \text{ m/s}$ and we performed a Monte Carlo analysis with 1000 runs to estimate the probability of detection (P_d), assuming a probability of false alarm $P_{fa} = 10^{-4}$. Figure 5 shows the estimated probability of detection as a function of the target signal-to-noise ratio (SNR), measured at the input of the overall processing chain. As expected, the longer arrays, namely the λ -ULA and the NULA, allow to detect the slower target for lower value of the input SNR. On the other hand, the $\lambda/2$ -ULA is attenuating this target due to the presence of a larger cancellation notch.

For a more comprehensive analysis, the detection performance was also evaluated for different target velocities. Figure 6 shows the P_d as a function of the target bistatic radial velocity, for a fixed input SNR of -45 dB . The results confirm the above considerations and lead to the following remarks:

1. For the $\lambda/2$ -ULA, we can clearly distinguish a larger cancellation notch, which is responsible for the attenuation of very slow-moving targets.
2. For the λ -ULA, the spatial undersampling leads to clutter folding and therefore to replicas of the cancellation notch in the velocity range of interest. These are responsible for the attenuation of faster targets.
3. Finally, the NULA shows a certain robustness against both the mentioned issues, enabling better detection performance for both slower and faster targets. This is obtained at the expense of some performance fluctuations at intermediate velocity values.

B. DoA estimation

After a target is detected, a fine ML estimation of the target DoA can be achieved by (8).

In practical applications, target localization is achieved in two steps. First, using a bank of filters spaced in angle by the expected beamwidth (BW), the radar searches for targets in the angular sector of interest. When a target is detected by one of the filters, a coarse estimate of the DoA is available. Then, the estimate can be refined via the ML estimation approach described. Therefore, in the following, we assume that the detected target is included in the main beam, and we limit the angular sector on which the ML estimator is applied to an extent of one BW.

Figure 6 (a) reports the DoA estimation accuracy as a function of the input SNR, for a slow-moving target with $v_b = 4 \text{ m/s}$. The results are expressed in terms of Root Mean Square Error (RMSE). For a more comprehensive analysis, the DoA estimation performance was also evaluated for different target velocities. Figure 6 (b) shows the resulting RMSE as a function of the radial velocity, for a fixed SNR of -45 dB . To provide a reference for the expected DoA estimation accuracy and better evaluate the impact of the considered array configurations, the corresponding Cramér–Rao bound (CRB) is also reported.

The results shown in Figure 6 lead to the following remarks:

- There is a good match between the theoretical results and the simulations, especially for high SNR values. This confirms the asymptotic efficiency of the ML estimator.
- As a general result, longer arrays yield a narrower main beam. Therefore, for most of the target velocities, both the λ -ULA and the NULA achieve a lower estimation error than the $\lambda/2$ -ULA.
- The dependence of the RMSE on the target velocity can be explained due to the different degree of attenuation obtained by the three arrays at different target velocities. Therefore, while the $\lambda/2$ -ULA shows a monotonic RMSE, the λ -ULA suffers from the presence of a blind velocity at $v_b = v_p$. Finally, the NULA shows a fluctuating RMSE at intermediate velocity values.
- For slower targets, the improved RMSE obtained by longer arrays is also the result of a better preservation of the target echo.
- For faster targets, the degraded detection performance of the λ -ULA leads to higher RMSE values, despite its narrower main beam. The estimation accuracy becomes even worse than that of the $\lambda/2$ -ULA for $v_b > 10 \text{ m/s}$.

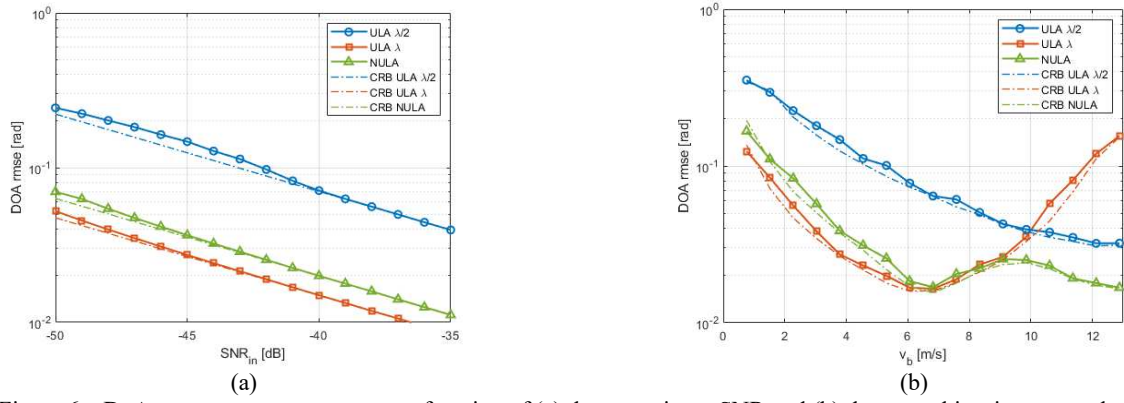


Figure 6 – DoA root mean square error as a function of (a) the target input SNR and (b) the target bistatic target velocity.

• Finally, the considered NULA solution allows to improve the estimation accuracy for both slower and faster targets. The observable fluctuation in the RMSE for intermediate target velocities corresponds to those also visible in the P_d curve. Nevertheless, the achieved accuracy is always better than that provided by the $\lambda/2$ -ULA.

IV. CONCLUSION

In this paper, we investigated the advantages of NULA for target detection and DoA estimation against clutter in passive radar systems mounted on moving platforms. We first remarked how, in passive radar systems typically employing a limited number of receiving channels, connected to antenna elements with low directivity, the requirements of accurate DoA estimation and wide unambiguous angular sector, as well as the range of interest of detectable velocities, may set competing constraints on the array configuration. In a ULA the need for a wide unambiguous angular sector limits the inter-element spacing. If a small number of elements is employed, this results in arrays of reduced length and, in turn, in a poor estimation accuracy. Furthermore, we remarked how using arrays of reduced length also undermines the system's ability to detect slow-moving targets, which might be cancelled along with the clutter. Conversely, if the total length of the array is increased to improve the detection of slow-moving targets and reduce the DoA estimation errors, both the presence of blind velocities and ambiguities in DoA estimation must be accepted.

With this in mind, we showed how the above limitations can be overcome by resorting to a non-uniform arrangement of the available antenna elements. Hence, we showed an overview of NULA potentialities in terms of target detection and DoA estimation, by considering the case of a mobile passive radar system exploiting DVB-T as a signal of opportunity and based on a STAP scheme.

In terms of target detection, we showed via simulated analysis how NULA configurations allow a better preservation of both slow and fast-moving targets, when compared to equivalent ULA configurations, by capitalizing on their spatial diversity. In terms of DoA estimation, we remarked that NULAs allow to obtain better estimation accuracies, comparable to those of an ambiguous λ -ULA, due to a narrower main beam.

The results obtained suggest that NULA configurations deserve more in-depth studies, and they are particularly

suitable for low-cost radar systems, such as passive radars, where the number of receiving channels must be kept low to satisfy compactness and lightness requirements, and where an accurate estimation of target DoA is a critical need. The reliability and the performance of the system were shown to be improved by exploiting NULA spatial diversity. In this perspective, a specific optimization strategy for the design of the NULA, considering clutter cancellation, target detection and DoA estimation requirements, could be the subject of future research.

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